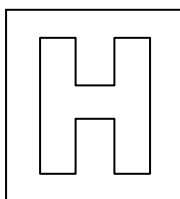


Candidate Name: _____

Class Adm No

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2016 Preliminary Examination II Pre-University 3

H2 Biology

9648/02

Paper 2 Core Paper

16 September 2016

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This question paper consists of 28 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1. Figure 1.1 shows a mid-section of a mammalian DNA molecule, comprising Strand **A** and Strand **B**, during the S phase of interphase.

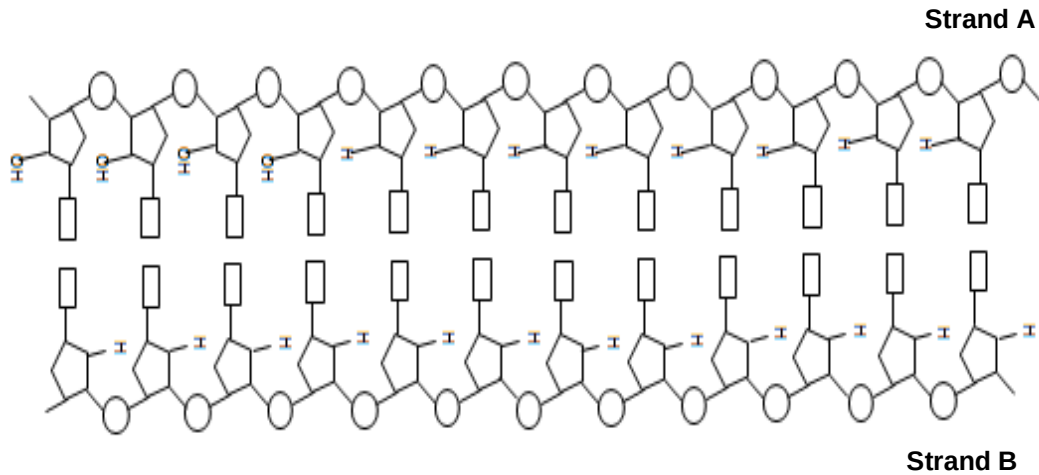


Figure 1.1

- (a) With reference to Figure 1.1,
- (i) account for the difference in the nucleotides in Strand **A** compared to Strand **B**.

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..... [3]

- (ii) Outline the next immediate step in the further processing of Strand **A**.

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..... [1]

Figure 1.2 depicts a telomere complex at the end of a chromosome in a human zygote. Telomeres are lengthened during embryonic development to compensate for shortening due to the end-replication problem.

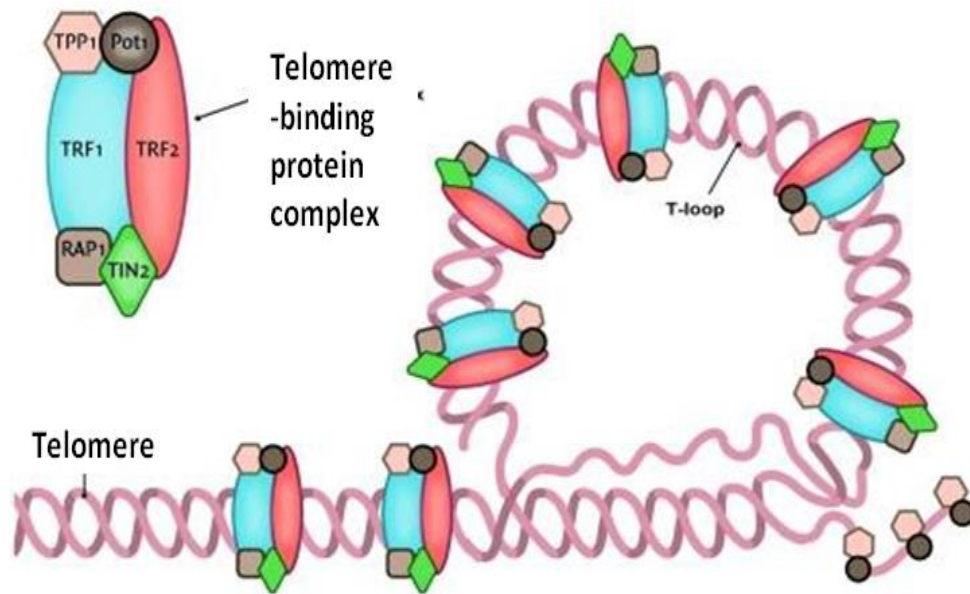


Figure 1.2

- (b) Describe two differences between transcription and the process of telomere lengthening.

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..... [2]

A study was conducted to investigate gene expression in yeast. Yeast strain **X** and strain **Y** were used in this study. These strains differed in the location of the *ADE2* gene. In strain **X**, *ADE2* gene is at its normal locus on chromosome 5. An engineered inversion event is responsible for creating strain **Y** where *ADE2* gene locus is near the telomere. Figure 1.3 depicts the loci of the *ADE2* gene in both strain **X** and strain **Y**.

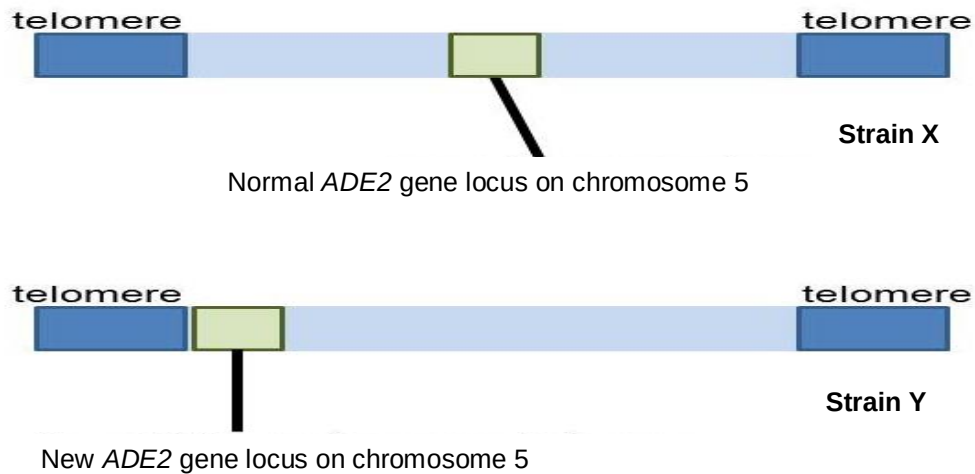


Figure 1.3

The above yeast strains were cultured on specially supplemented agar plates and incubated for one week to produce yeast colonies. Based on the constituents of the agar, expression of *ADE2* gene prevents accumulation of red pigments and results in cells being white in colour. The colonies produced from this period of incubation are represented in Figure 1.4.

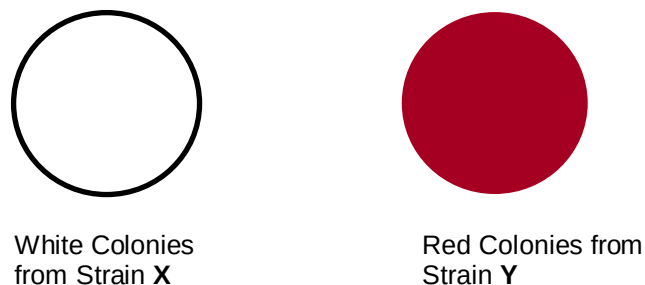


Figure 1.4

Previous studies have shown that some proteins bind to yeast telomeres to recruit histone deacetylase enzymes. These enzymes then act on the chromatin regions around the telomeres.

- (c) Using the above information, explain the difference in colony colour between strain X and strain Y after one week of incubation.

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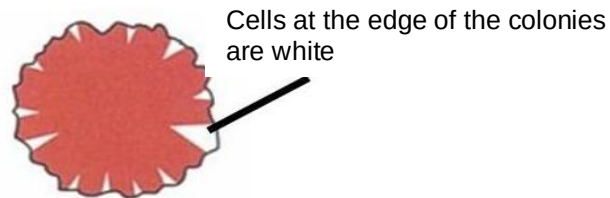
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..... [4]

The colonies from both strains were allowed incubate for an additional week. Following this, it is then observed that some cells at the edges of the Strain Y colonies appeared white in colour, as depicted in Figure 1.5.



Appearance of colonies from Strain Y after two weeks of incubation

Figure 1.5

- (d) Explain why the cells at the edges of Strain Y colonies appear white in colour after two weeks of incubation.

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..... [3]

[Total: 13]

2. In the cytosol, chaperone proteins can bind reversibly to newly synthesized polypeptides and shield them from other molecules. This prevents the R groups of amino acids from forming bonds with the wrong molecular partners, hence allowing the polypeptides to fold correctly.

Figure 2.1 below shows how chaperone proteins work.

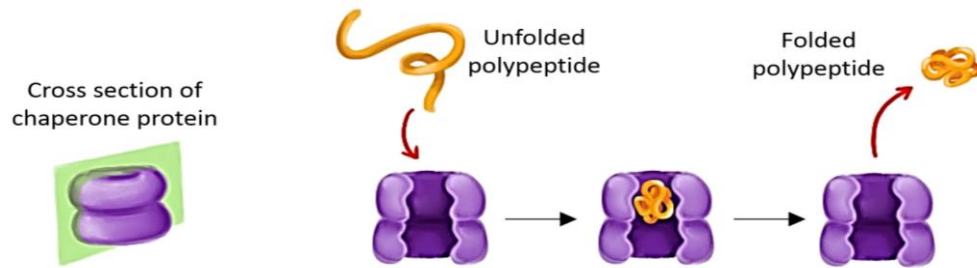


Figure 2.1

- (a) With reference to Figure 2.1, explain how chaperone proteins regulate gene expression.

.....

 [2]

- (b) Describe how cytosolic proteins are degraded in cells.

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 [2]

When a protein fails to fold correctly upon synthesis or misfolds at a later stage in its cellular life time, it can no longer fulfil its biological function. Protein misfolding may occur due to mutations in genes or interference with expression of chaperone proteins. Figure 2.2 shows how misfolded proteins may aggregate in a cell.



Figure 2.2

- (c) With reference to Fig. 2.2, suggest why misfolded proteins tend to aggregate in the cell.

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..... [2]

An example of a misfolding disease is Bovine spongiform encephalopathy (BSE), a neurodegenerative disease in cattle and commonly referred to as mad cow disease. In BSE, there is an aggregation of misfolded prion proteins in the neurons of cattle, which results in degeneration of the brain and spinal cord.

Hereditary forms of BSE have been associated with a mutation in the brain prion protein (PrP) gene, which encodes the PrP protein. The PrP protein is a transmembrane glycoprotein found on the surface of neurons. Figure 2.3 shows the ribbon model of a normal PrP protein.

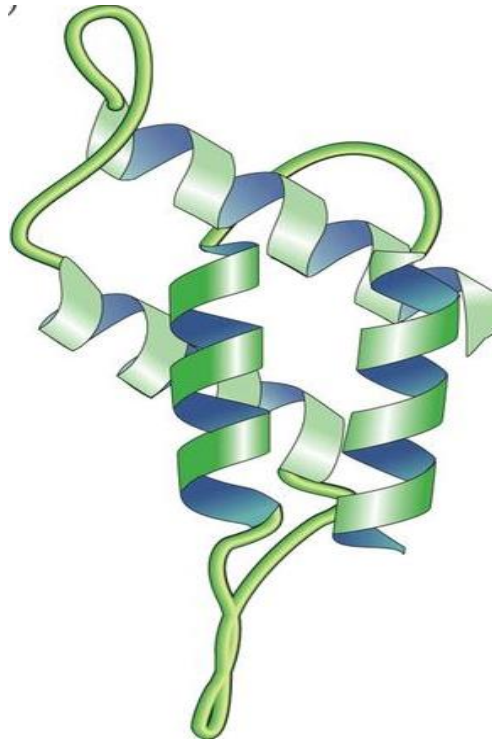


Figure 2.3

- (d) Explain how the the PrP protein would maintain its position within the cell membrane.

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..... [2]

- (e) Describe how the PrP protein on the cell surface membrane is formed after its polypeptide chain has been synthesized.

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..... [4]

A human version of BSE called variant Creutzfeldt-Jakob disease (vCJD) is believed to be caused by eating beef products contaminated with central nervous system tissue from infected cattle. Studies have shown that the introduction of misfolded cattle PrP by consuming contaminated meat can cause normal human prion protein to misfold and form aggregated plaques in the brain.

Health advisories have cautioned that cooking contaminated meat will not provide protection from infection that would lead to vCJD.

- (f) Suggest why cooking contaminated meat products will not provide protection from contracting vCJD.

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..... [1]

[Total: 13]

3. Burmese cats, *Felis catus*, show discontinuous variation in the colour of their eyes and hearing ability. A large scale investigation was conducted to study the inheritance pattern of eye colour and deafness in such cats.

Pure-breeding blue eyed cats which were deaf were crossed with pure-breeding yellow eyed cats with normal hearing.

The F_1 all had yellow eyes and normal hearing.

Female cats from the F_1 were crossed with male cats with blue eyes and were deaf. 668 offspring were produced.

The observed numbers of F_2 cats with each phenotype were as follows:

Blue eyes, deaf	259
Blue eyes, normal hearing	72
Yellow eyes, deaf	53
Yellow eyes, normal hearing	284

- (a) State what is meant by pure-bred cats for eye colour and hearing ability.

.....
 [1]

A chi-squared test, with the guide of Table 3.1, was planned to evaluate the difference between observed and expected results.

Table 3.1

Degree of Freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (b) Based on this study, state the expected results for F_2 cats produced from crossing F_1 female cats with male cats with blue eyes and were deaf.

..... [1]

- (c) With reference to Table 3.1, outline how the significance of the difference between observed and expected results can be evaluated with 98% confidence (Formulae not required).

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..... [2]

The difference between observed and expected results was eventually concluded to be significant.

- (d) Using the symbols Y/y for eye colour and H/h for hearing ability, draw a genetic diagram to explain the observed results of the cross between F₁ female cats and male cats with blue eyes and were deaf.

[4]

The coordinated activity of higher order organisms like cats relies upon a continuous input of information from the internal and external environments. Information is in the form of stimuli, which are detected by receptors. One such receptor is called a Pacinian corpuscle, which is found in the skin.

Fig. 3.1 shows the electrical activity recorded by two microelectrodes inserted into:

- the axon within the Pacinian corpuscle at site **P**, and
- the axon of the sensory neurone leaving the corpuscle at site **Q**

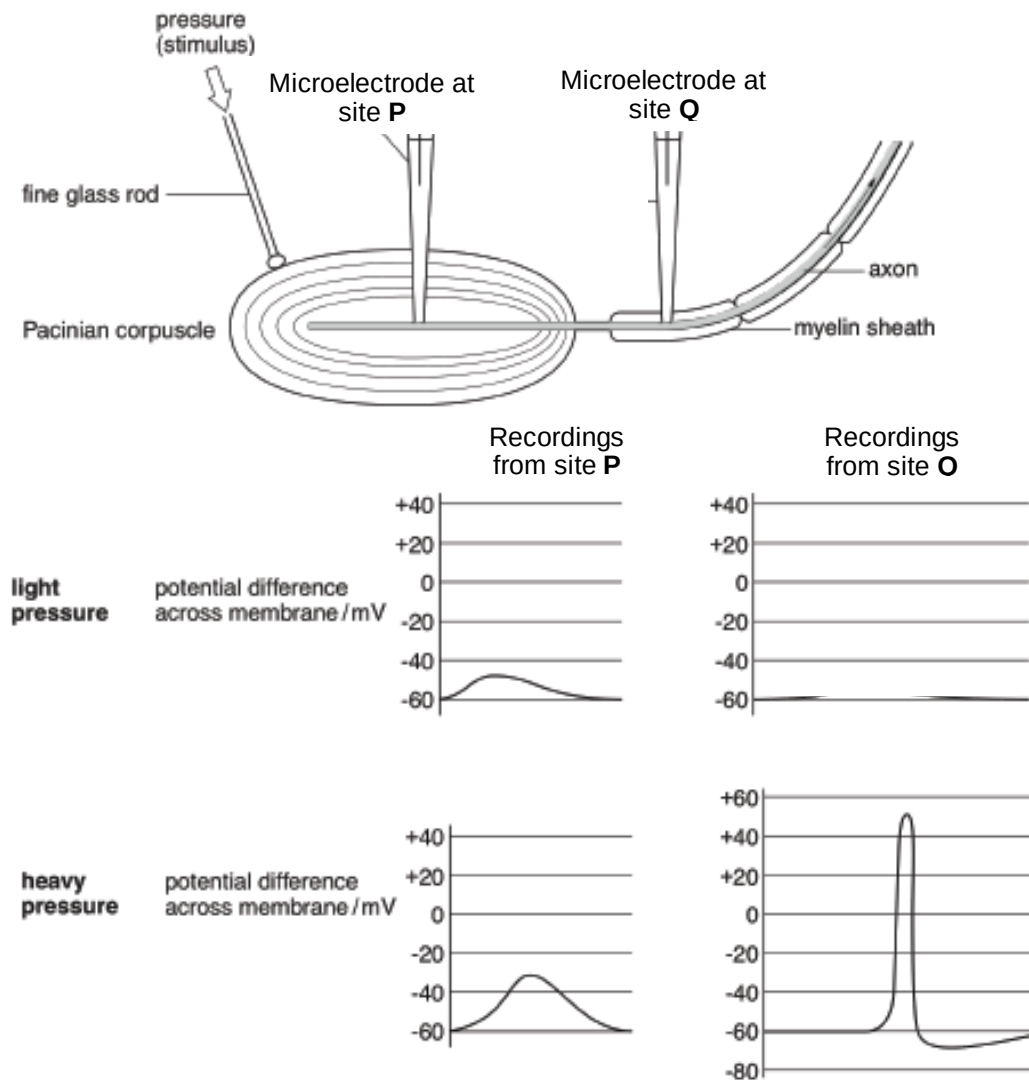


Figure 3.1

- (e) With reference to Figure 3.1, account for the changes in electrical activity in site **P** and **Q** as increasing pressure is applied to the Pacinian corpuscle.

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..... [4]

[Total: 12]

4. Rhinoviruses are the most common viral infectious agents in humans and are the predominant cause of the common cold. Symptoms include sore throat, runny nasal congestion, sneezing and cough which may be accompanied by muscle aches and loss of appetite.

The structure of the human rhinovirus is depicted in Figure 4.1. Human rhinoviruses are composed of a capsid that contains four viral proteins **VP1**, **VP2**, **VP3** and **VP4**. Each virion has one copy of single-stranded positive sense RNA genome of between 7200 and 8500 nucleotides in length.

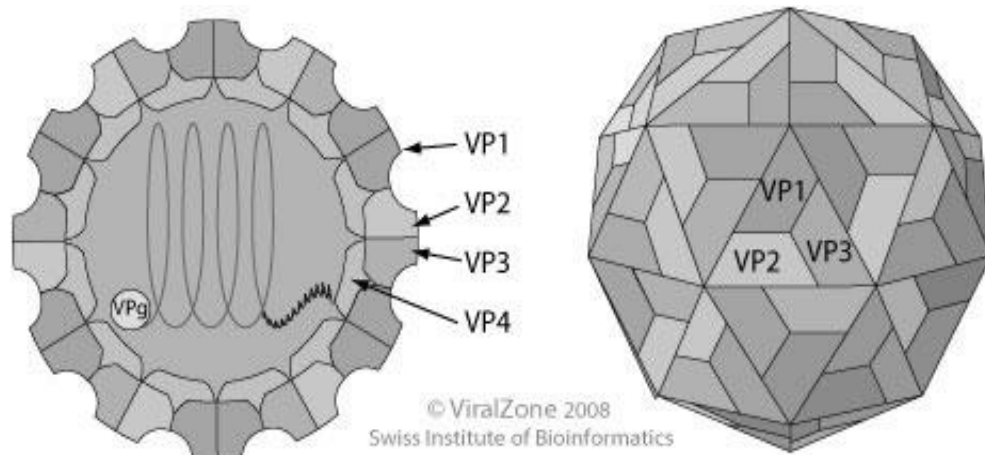


Figure 4.1

Despite structural differences, both human rhinovirus and influenza virus are able to penetrate respiratory epithelial cells for infection.

- (a) Explain how it is possible for both human rhinovirus and influenza virus to be able to penetrate human respiratory epithelial cells.

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 [1]

The enzyme neuraminidase plays an important role after new influenza virions exit their host cells from each round of infection.

- (b) Describe the role of neuraminidase in the reproductive cycle of the influenza virus.

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 [2]

- (c) With reference to Figure 4.1, explain why it is unlikely that the newly assembled human rhinovirus exit their host cells in the same way as influenza virions.

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..... [2]

H7N9 is a subtype of Avian Influenza virus, which normally circulates amongst the avian populations with some variants known to occasionally infect humans. An H7N9 virus was first reported to have infected humans in 2013 in China and now this virus has spread among humans. Figure 4.2 shows the proposed evolution of H7N9 virus.

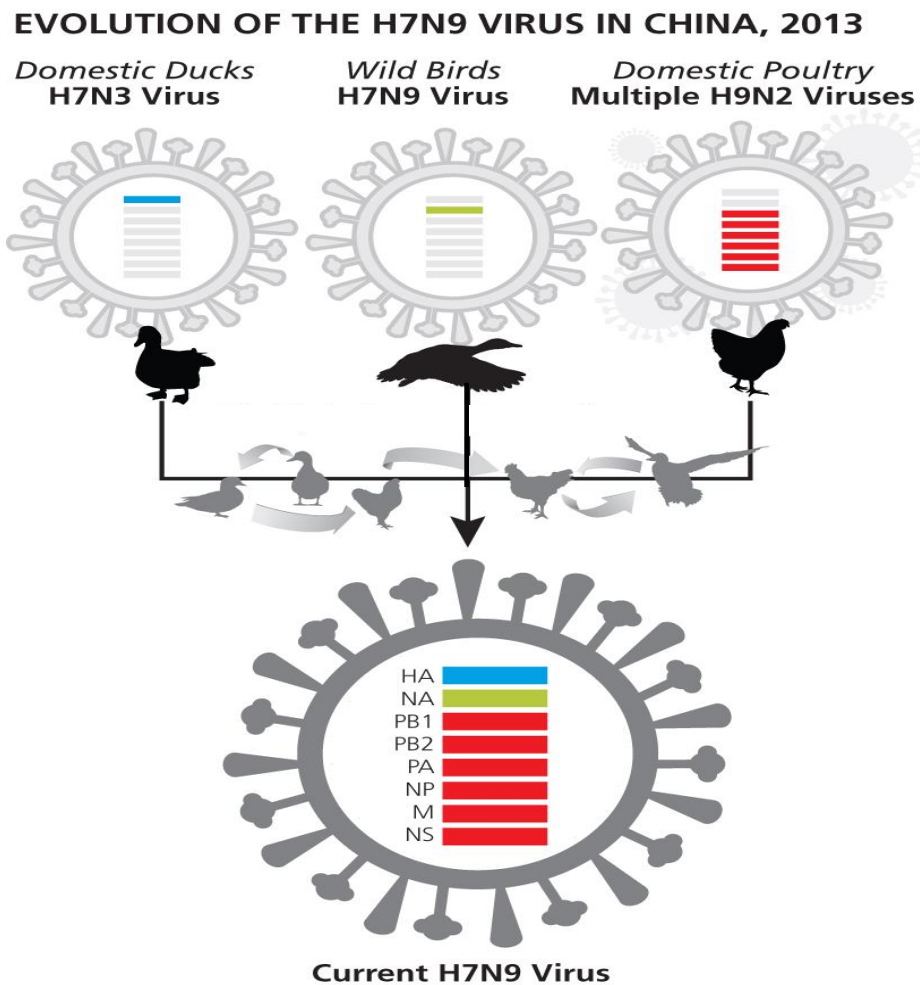


Figure 4.2

- (d) With reference to Figure 4.2, explain how the new combination of RNA segments in H7N9 was proposed to have arisen.

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..... [3]

Today viruses are considered an exception to the cell theory which states that the basic units of life are cells. Viruses have been referred to as “organisms at the edge of life”.

(e) State one characteristic of viruses that may classify them as being

(i) living

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..... [1]

(ii) non-living

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..... [1]

[Total: 10]

5. A study was carried out to measure the concentrations of glucose and insulin in the blood of students. The results are summarised in Fig. 5.1.

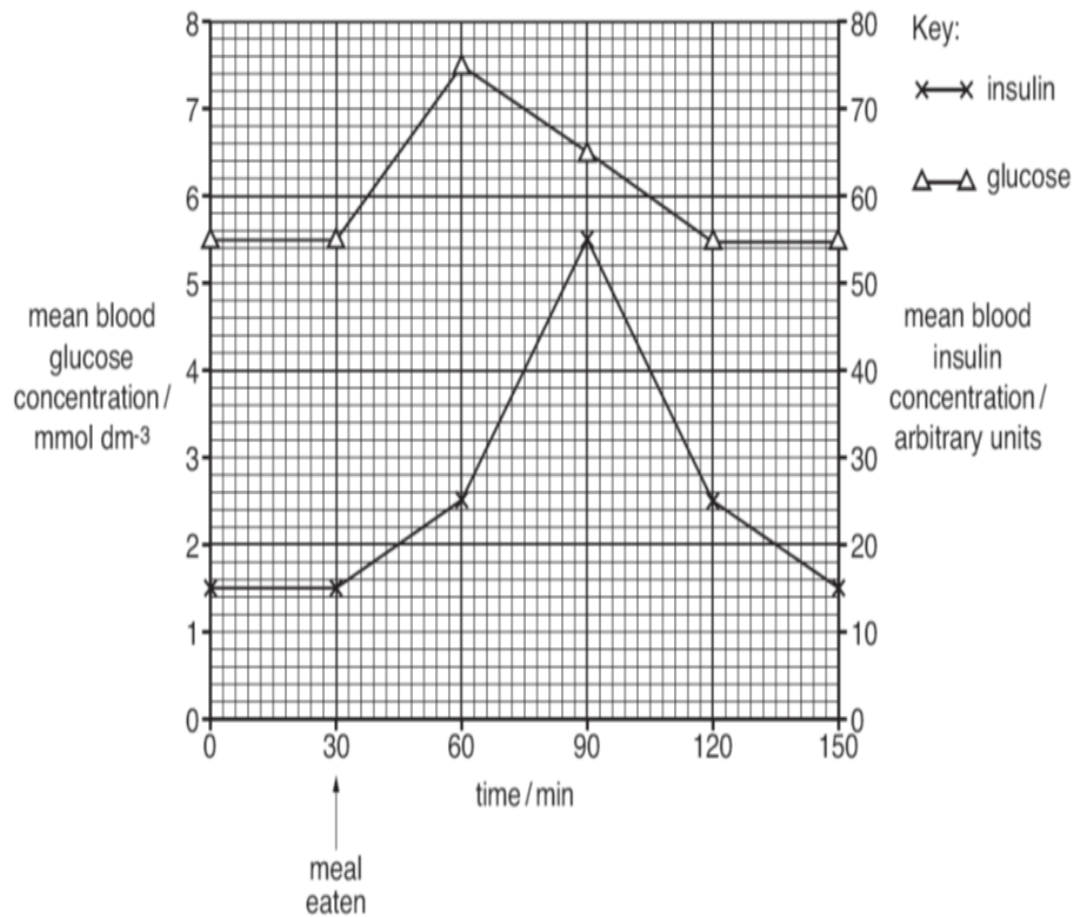


Figure 5.1

- (a) With reference to Figure 5.1, account for the relationship between the concentration of glucose and the concentration of insulin in the blood after a meal.

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..... [5]

Figure 5.2 outlines some steps in glucose metabolism in human muscle cells.

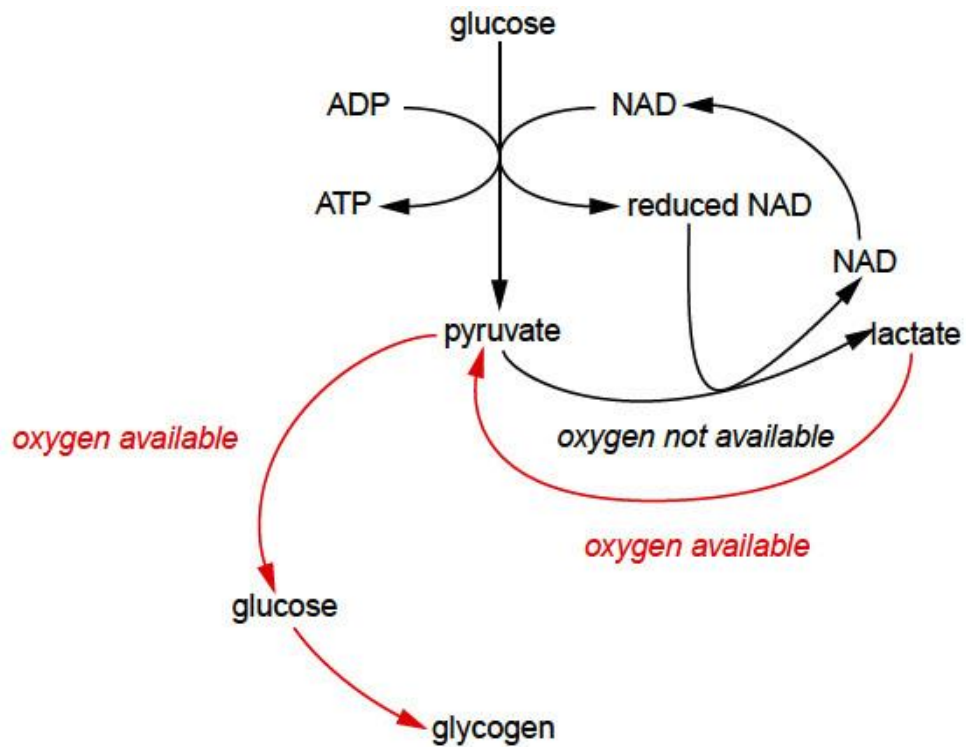


Figure 5.2

- (b) Explain why pyruvate needs to be converted to lactate in the absence of oxygen.

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..... [2]

Hexokinase is an enzyme that plays a critical role in glucose metabolism. This enzyme converts glucose into glucose-6-phosphate. This step is important as it energises glucose for further metabolic reactions. Fig. 5.3 is a computer-generated image of the enzyme hexokinase binding with its substrate, glucose.

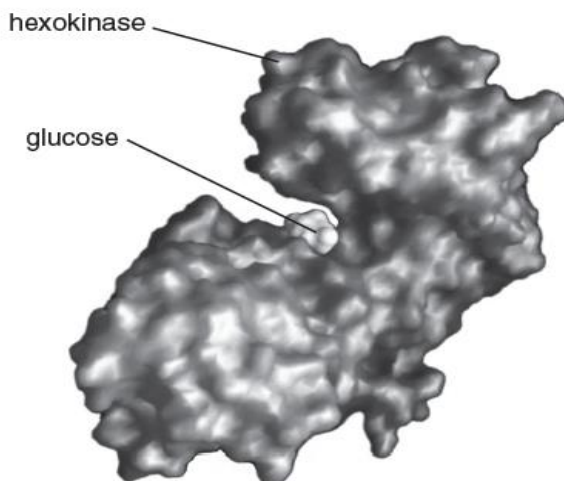


Figure 5.3

Crystallization studies have shown that the shape of hexokinase's active site is not complementary to the shape of glucose.

(c) Describe the mechanism by which hexokinase binds to glucose.

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..... [2]

The phosphorylation of glucose by hexokinase also prevents it from leaving the liver cells.

(d) Suggest why glucose-6-phosphate cannot move out of liver cells.

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..... [2]

[Total: 11]

6. Most ATP is made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another. Figure 6.1 show two such organelles.

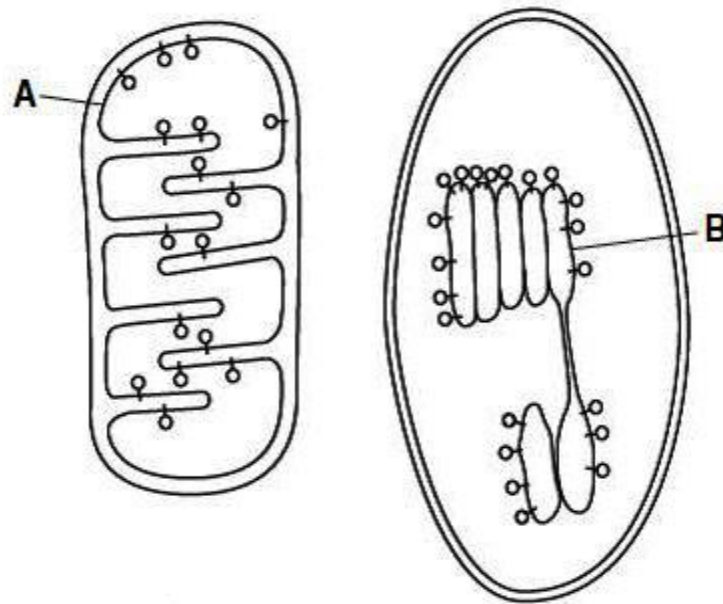


Figure 6.1

- (a) Draw arrows onto each organelle in Figure 6.1 to show the direction in which protons are pumped.

[2]

- (b) In addition to facilitating the formation of proton gradients, state one other importance of compartmentalisation within organelles **A** or **B** for ATP synthesis.

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..... [1]

In 1882, the German botanist T.W. Engelmann performed an experiment to investigate the effects of different wavelengths of light on the rate of photosynthesis using *Spirogyra*. *Spirogyra* is a type of green alga that contains photosynthetic organelles. Other studies have shown that there are pigments that absorb blue, green, yellow and red light in these organelles.

A strand of *Spirogyra* was placed into water containing aerobic bacteria. Different parts of the strand were exposed to different colours of light. After a period of time, the bacteria had moved into the positions shown in Figure 6.2.

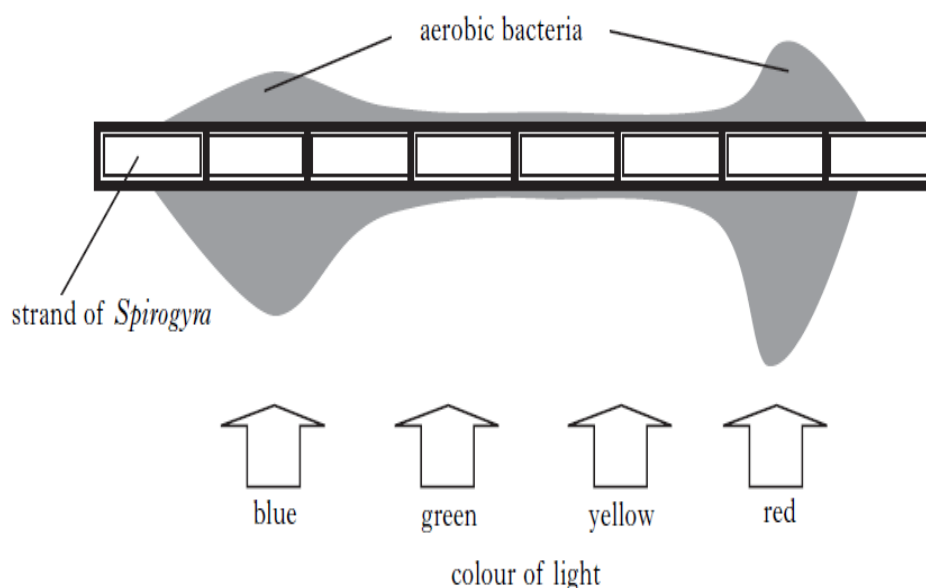


Figure 6.2

It was concluded that wavelengths corresponding to red and blue light caused a higher rate of photosynthesis compared to wavelengths corresponding to green and yellow light.

(c) With reference to Figure 6.2,

(i) explain why the abovementioned conclusion was drawn.

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..... [3]

- (ii) suggest a role for the pigments absorbing green and yellow light in Spirogyra's photosynthetic organelles.

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..... [2]

[Total: 8]

7. Wolf, *Canis lupus*, is one of the world's best known and well researched species, with arguably more books written about it than any other wildlife species. It is the largest extant member of its family, *Canidae*, with males averaging 45kg and females 37kg.

Wolves are further classified into a number of sub-species based on marked phenotypic differences, such as body size and colour. Figure 7.1 shows the several sub-species of wolves in existence today.

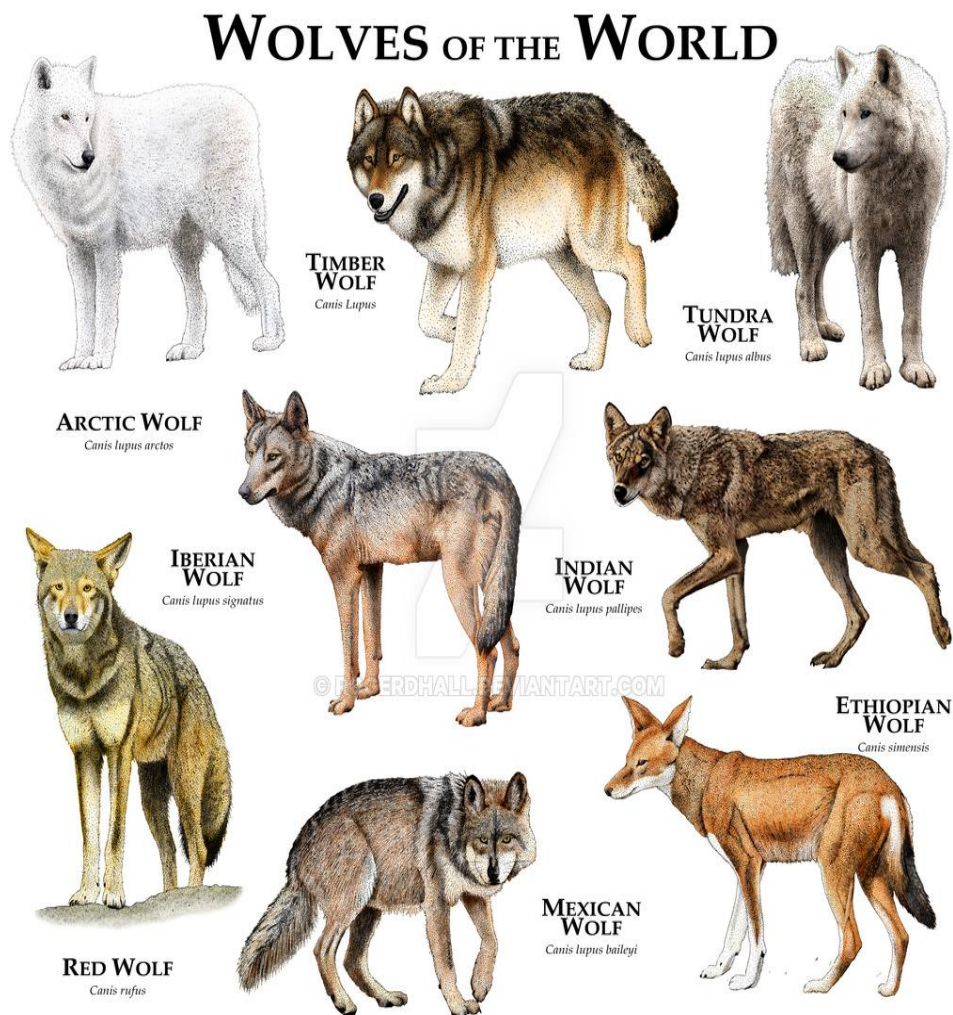


Figure 7.1

- (a) Suggest one reason why the existing populations of wolves are classified as different sub-species rather than different species.

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 [1]

Timber wolves are native to the wilderness of Eurasia and North America. Studies have suggested that the timber wolves once co-existed with the extinct dire wolf, *Canis dirus*, on the North American continent.

Figure 7.2 depicts skeletal models of a timber and dire wolf. Dire wolves were larger and had more powerful jaws compared to timber wolves. However, they had proportionally much shorter legs than timber wolves, and were as such slower runners. They are known to feed on different prey, with the large dire wolves favouring larger prey and the timber wolves favouring smaller prey. It is known that the dire wolves went extinct approximately 8000 years ago, which coincides the time period when its large prey declined in numbers.



Figure 7.2

- (b)** Using Darwin's theory of evolution, explain why the dire wolves went extinct.

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..... [3]

- (c)** Explain why a population is the smallest unit that can evolve.

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..... [2]

Scientists used DNA hybridisation to determine the evolutionary relationships between five members of the *Canidae* family. The temperature at which a molecule of double-stranded DNA separates into two single strands is the separation temperature. A region of non-coding DNA was analysed for the study. The scientists recorded the mean separation temperature of this region of DNA in which both strands were from the same species. The scientists then recorded the mean decrease in separation temperature of DNA in which one of the strands was from another species. Their results are shown in Table 7.1

Table 7.1

	Mean decrease in separation temperature/ °C				
	Timber wolf	Dog	Coyote	Ethiopian wolf	Dhole
Timber wolf					
Dog	1.7				
Coyote	2.3	2.3			
Ethiopian wolf	3.6	3.6	3.5		
Dhole	4.8	4.8	4.7	4.9	

- (d) With reference to Table 7.1, explain why Dholes are most distantly related to the timber wolf based on the *Canidae* members studied.

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..... [3]

The scientists assume that the decreases in separation temperatures are directly proportional to the time since the evolutionary lines of Canidae members separated. Dogs are thought to have separated from timber wolves 40000 years ago.

- (e) Using this information, calculate how long ago, to the nearest year, Ethiopian wolves and timber wolves separated. Show your working.

[2]

- (f) Distinguish classification from phylogeny.

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..... [2]

[Total: 13]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
 Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
 Your answers must be in continuous prose, where appropriate.
 Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8.
- (a) Compare and contrast the reproductive cycles of lambda phage and HIV. [8]
 - (b) Describe the structural features and regulation of the tryptophan operon. [7]
 - (c) Explain the role of geographical and behavioural isolating mechanisms in the evolution of new species [5]

[Total: 20]

- 9.
- (a) Compare and contrast triglyceride and starch as a storage molecule. [6]
 - (b) Explain why cancer is a multi-step disease. [8]
 - (c) Describe the role of receptor tyrosine kinases in insulin signalling. [6]

[Total: 20]

End Of Paper